Subject: INTRODUCTION TO ELECTRONIS ENGINEERING.

Subject Code: 22ESC143

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Module 5

Chapter	Title
1.	Analog Communication Schemes
2.	Digital Modulation Schemes.

Model Question

1. Write a note on different types of modulation schemes & Briefly describe each in detail. (8M)

2. Brief about modern Communication S/m with its block diagram. (7M)

3. List out the advantages of Digital communication over analog communications. (5M).

4. With its neat diagram, explain the concept of Radio Wave propagation & its dif^{nt} types, (7M)

5. Describe Radio Signal Txn & Multiple access techniques. (7M)

6. Consider the following binary data & sketch the ASK, FSK, PSK modulated w/f. (6M).

COMMUNICATION

- Communication is the transfer of information from one point to another point (or) one person to another person (or) one place to another place.
- The information to be transferred may be in the form of text, data, voice, sound, videos, images etc.
- The information present in any of the form can be transmitted from source to destination, depending on the distance through wired (or) wireless form.
- The communication was divided as two types,

→ Analog communication, } Depending on the form of signal transmission in the transmission medium.
→ Digital communication.

& they have divided as:

→ Wired Communication, } Depending on the
→ wireless —"—. } type of transmission physical medium.

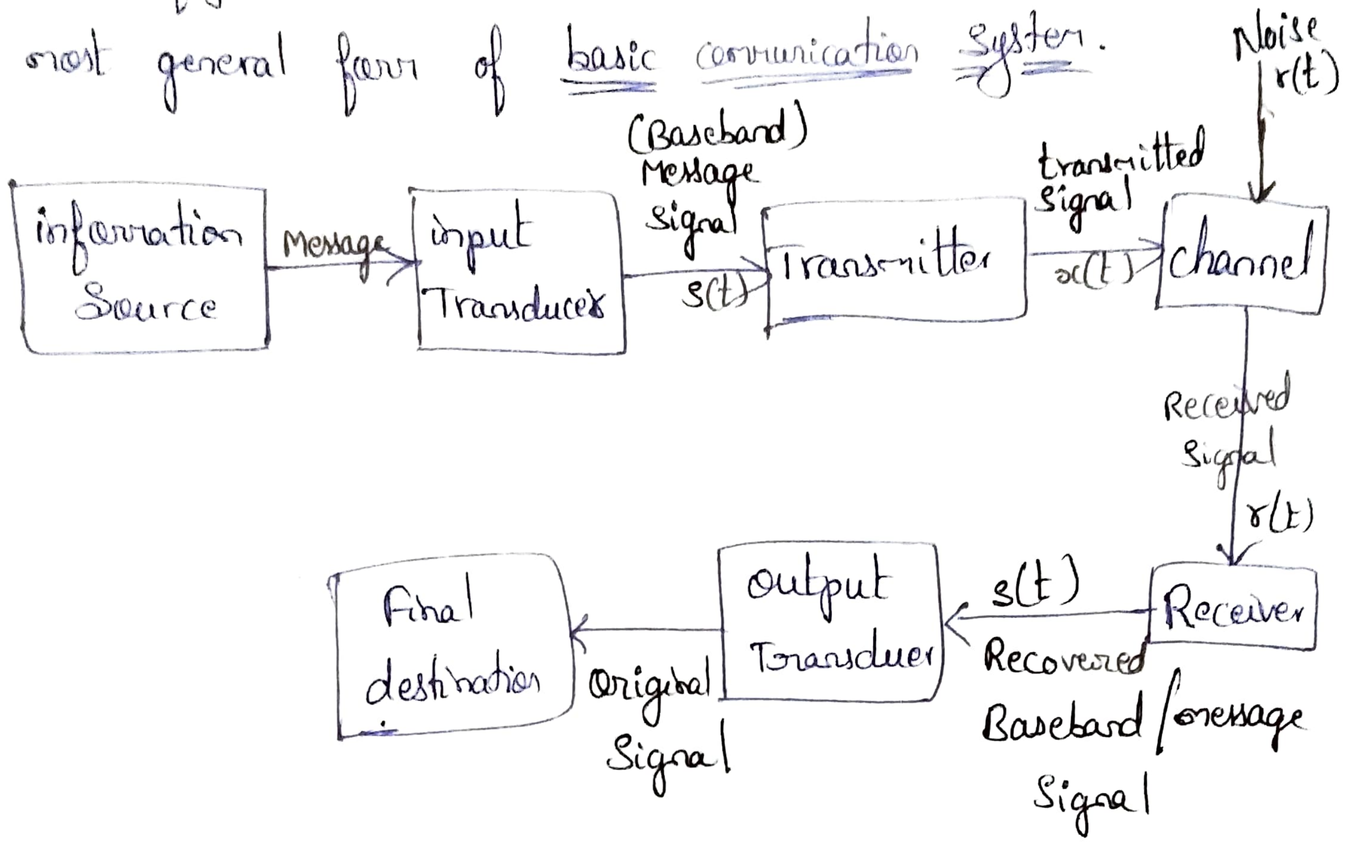
(7M) MODERN COMMUNICATION SYSTEM SCHEMES :

*** (Brief about the modern communication S/m & its block diagram.)

* Modern communication is the science & practice of transmitting information through electrical signal which is referred as communication Engineering.

* In a Communication S/m, the information from a Source is reaches the destination only through number of intermediate Subsystems (or) stages Etc.

* Below figure shows a block schematic diagram of the most general form of basic communication system.



↝ The main constituents (or) blocks (or) subsystems of basic communication system are :

1. Information Source,
2. Input transducer,
3. Transmitter,
4. channel or medium,
5. Noise,
6. Receiver,
7. output transducer,
8. Final destination,

* There are different types of communication systems, like analog, digital, radio Etc. But, each of those have general communication system blocks as shown above, but with different principle of operation.

* Consider now each of the block explanation.

1. Information Source:

* In a communication s/m, the information (or) message which is to be transmitted from Transmitter to receiver, is taken from information Source.

For Ex: that information may be present in the form of text, voice/sound, images, Video's Etc, for which the Source is a "person".

2. Transducer: —

* Transducer is a device, which converts one form of energy into another form.

Input transducer: — is a device, which converts non-electrical signal energy into electrical signal.

output transducer: — is a device, which converts electrical signal into non-electrical signal form.

Ex: Some examples of transducers are:

- Microphone (converts sound ~~electrical~~ signals to electrical signals),
- TV picture tube converts electrical to corresponding pictures.
- Movie cameras, Video casettes, Recorders, Etc.

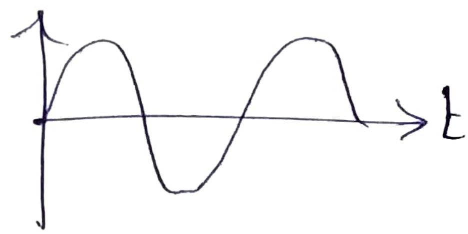
* The information produced by the information source is applied to the i/p transducer, which converts that into electrical signal & that electrical signal is referred as Base band signal (or) Message signal or information signal or an intelligent signal or an envelope, which is represented as s(t) in communication theory.

* There are two types of signals. They are :
  - Analog Signals,
  - Digital —"—.

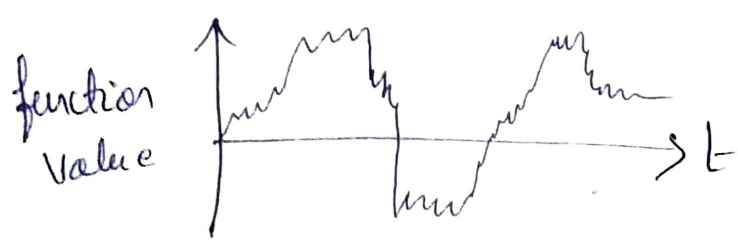
Analog Signals : —

* An analog signal is a function of time & has a continuous range of values.

Ex : (i) Sine wave



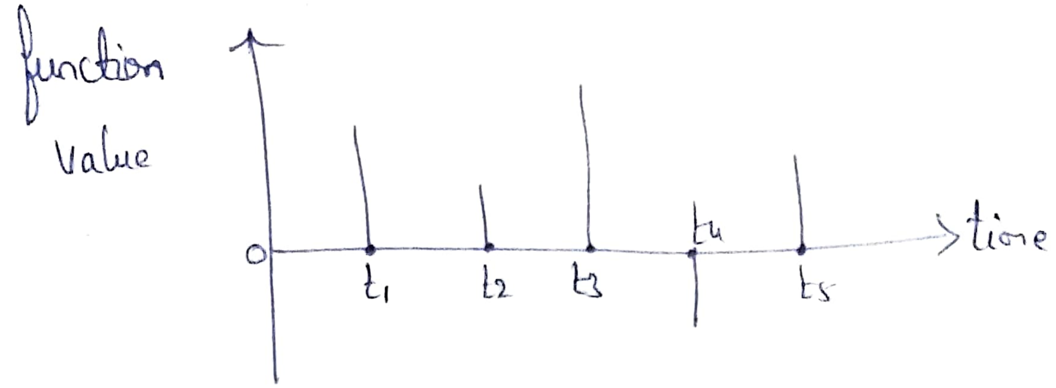
(2) Practical example is Voice signal.



Digital Signal : —

* Digital Signals are discrete in nature, they are not continuous with time function.

* The example of a digital Signal is shown below.



ADC [Analog to Digital Converter]:

* ADC converts an analog Signal into a digital Signal, by using the process of:
  - → Quantization,
  - → Sampling.

Quantization:—

→ Quantization is the process of converting continuous amplitude of an analog signal into discrete amplitude.

→ Sampling → is the process of converting a Signal which is continuous with time into discrete time.

→ The quantization & Sampling processes together forms Analog - to - Digital conversion & the circuitry used to perform this operation is referred as ADC (Analog to Digital converter).

TRANSMITTER : — (TXR)

* In a communication s/m, after i/p information source, i/p Transducer, the next block is the Transmitter.

* The TXR receives baseband Signal S(t) from the i/p transducer part.

* Prior to transmission, the TXR processes the baseband Signal. The processing may be any of the type as shown below.

① The baseband signal with low frequency spectrum is translated into high frequency spectrum.

② The baseband signal transmitted without translating it to a high frequency spectrum.

* The processing of baseband signals prior txn is based on the type of communication system. i.e

→ Baseband communication system,

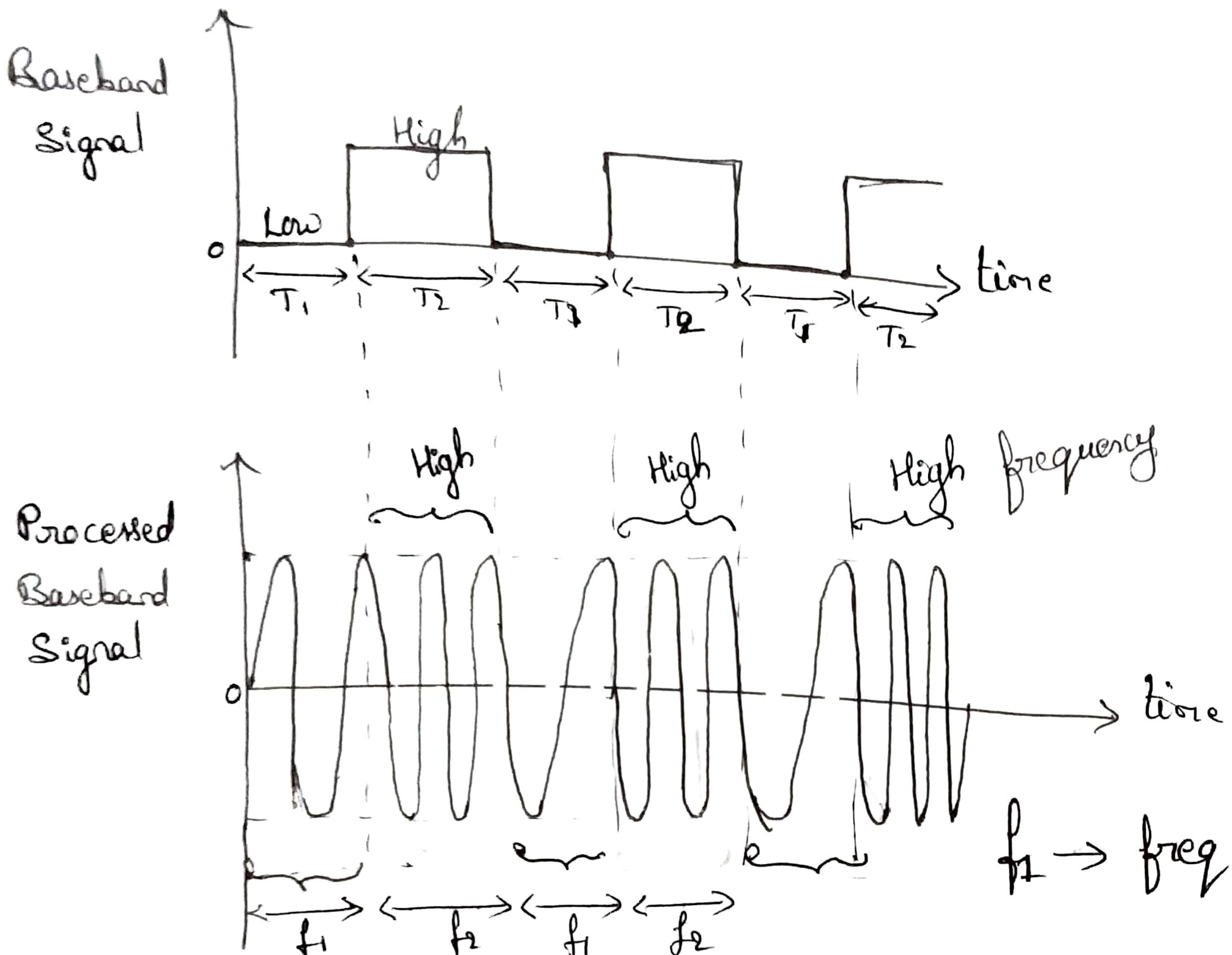
→ Carrier Communication system.

* In Baseband Communication System, the baseband signal is transmitted without translating it into a high-frequency spectrum signal.

* In carrier Communication system, the baseband signal is transmitted after translating it into a high frequency signal, which is referred as "Carrier Signal."

* The Signal is txd after processing as with translation or without translation also.

* For Ex:



$f_1 \rightarrow$ freq representing Low,

$f_2 \rightarrow$ freq representing High.

Thus, Baseband Signal is converted into a corresponding series of Sine waves of two different frequencies, prior to transmission.

* As in carrier communication s/m, the baseband Signal with low frequency is translated into High frequency spectrum, this process is referred as "Modulation".

→ Based on the type of baseband Signal in carrier communication system, the communication system is divided as analog or digital communication system.

→ If the baseband Signal is analog type of Signal, then the carrier communication system is referred as Analog Communication System.

→ If the baseband Signal is digital type, then the carrier communication s/m, is referred as Digital Communication System.

Block Diagram of a typical TXR :-

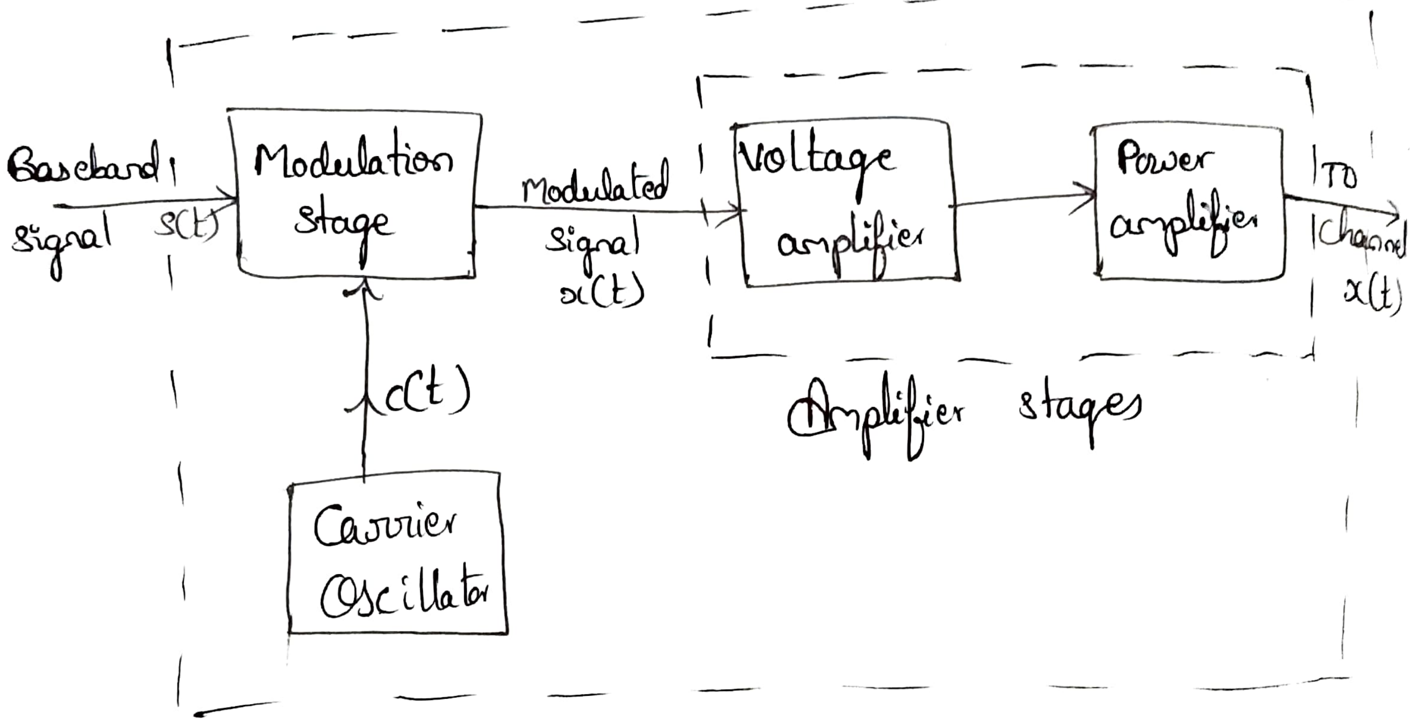
* A TXR performs some operations prior to transmission, they are

→ Modulation,

→ Amplification ⇒ Voltage amplification,

Power ——— " .

* The block diagram of a modern communication system (Transmitter Section) is shown below.



→ A baseband signal s(t) from the i/p transducer is applied to the modulation stage of the transmitter, where baseband signal with low frequency is translated into high frequency signal by adding with carrier signal c(t) of the oscillator. Then, x(t) modulated signal is obtained.

→ Now, the modulated signal $x(t)$ is applied to the amplifier stage of the transmitter, which has voltage amplification & power amplification to give strength to the signal to reach the destination.

→ Finally, from the transmitter the signal $x(t)$ is transmitted to the channel (or) medium.

CHANNEL (or) MEDIUM :—

* The channel is the transmission medium between transmitter & Receiver Section.
* During the transmission of a signal through the channel, there may be addition of external noise.
* The channel characteristics may affect the transmitting power, signal BW & cost of the communication s/m, as channel characteristics is also one of the design parameters of a communication system.

* Depending on the physical implementation, the channels are divided into two types. They are :

→ Hardware channels,

→ Software —"—.

* Hardware channels are manmade structures, which can be used as transmission medium.

For Ex: Transmission Lines, Waveguides, OFC (Optical Fiber Cables).

→ Transmission Lines are used only for Low (or) medium frequency signal transmission, but not for UHF signals.

Ex: Twisted pair cables ⇒ in Landline telephony,

Co-axial cables ⇒ in cable TV transmission.

(In antennas, Satellite communication).

→ Waveguides are used to transmit UHF range signals. (UHF → Ultra High frequency). These are hallow, circular (or) rectangular ~~wave like~~ metallic structures. The signals are reflected at the metallic wall (or) propagates to the other end of the waveguide.

→ OFC are highly sophisticated txn media, in which signals are transmitted in the form of light energy.

Ex: In telecommⁿ companies to txt telephone signals, Internet communicati

* In the communication s/m, if the channel is a physical link, then it is referred as Line (or) wired communication system.

Software channels :—

* If there is no any physical link exist in between TXR & RXR, then that channel is referred as Software channel & the communication system is referred as Radio Communication system.

* If the channel is an air (or) free space (or) a sea water, then the signal from transmitter is transmitted to software channel by converting that baseband modulated electrical signal into radio wave by using a "transmitting antenna" at the TXR block.

* Again at the Receiver, the radio signal must be converted into electrical signal by the receiving antenna.

//.

Ex: Radio Broadcast, Satellite communication, Cellular mobile communication } communication Systems in which Txn medium is Software channel.

NOISE :-

* Noise is an unwanted Signal, which adds with a Signal in the transmission medium or channel.
* It reduces the Signal Strength, which makes the signal not able to reach the actual destination. Hence, the power of the Signal is amplified in the TXR block in order to overcome from this problem.
* That is SNR (Signal to Noise Ratio) should be as high as possible.
* The addition of Noise in the transmission medium with the signal is referred as External (or) Extraneous Noise.
* If the Signal strength is varied because of variation in atoms of a component used in the S/m

(or) electronic variations, is referred as Internal Noise.

* ## External Noise :—

External Noise can be divided into 3-groups.

They are :

→ Atmospheric (or) static Noise,

→ Man-made (or) Industrial Noise,

→ Extra terristrial (or) Space Noise

Internal Noise :—

* The internal Noise can be grouped as :

→ Thermal or ~~shot~~ white noise, or Johnson's Noise,

→ shot or Transistor Noise.

* The thermal noise is generated due to the presence of resistors in the system circuits. As variations in the movement of

electrons or molecules in the resistor, the thermal noise will generate.

→ The maximum o/p noise power P_n of a resistor is given as:

$$\boxed{P_n \propto TB}$$

where,

T → Temperature (absolute)

B → Bandwidth.

$$\boxed{P_n = kTB}$$

where,

k → Boltzmann's constant.

$$\boxed{k = 1.38 \times 10^{-23} \ J/K}$$

* **Shot or Transistor Noise** :—

This noise is generated inside a "Transistor", because of random movement of electrons & holes inside the transistor.

If this noise with signal is amplified and fed to the speaker, it produces an effect similar to lead shot falling on a metallic sheet & hence the name is "Shot noise".

SNR [Signal to Noise Ratio] :—

* SNR is the ratio of the signal power to the noise power at ~~the~~ a point in the circuit.

$$\frac{S}{N} = \frac{P_s}{P_n} = \frac{V_s^2 R}{V_n^2 R}.$$

$$\boxed{\frac{S}{N} = \frac{P_s}{P_n} = \frac{V_s^2}{V_n^2}} \quad \& \quad \boxed{\left(\frac{S}{N}\right)_{dB} = 20 \log\left(\frac{V_s}{V_n}\right)}$$

where,

$P_s \rightarrow$ Signal power,

$P_n \rightarrow$ Noise —"—.

Noise Figure (F):— is a measure of the noise increased by the circuit. It is defined as the ratio of S/N at the i/p to the S/N at the o/p of the circuit.

Noise Measurement Technique :—

Noise measurement Technique :—

* Noise can be measured at any point in the circuit, which is referred as Noise figure.
* The device in which noise to be measured is ~~d~~ referred as Device Under Test (DUT).
* The Noise is measured by using a "Diode Noise Generator (DNG)".

→ The DUT is connected at the o/p of the Diode Noise Generator (DNG).

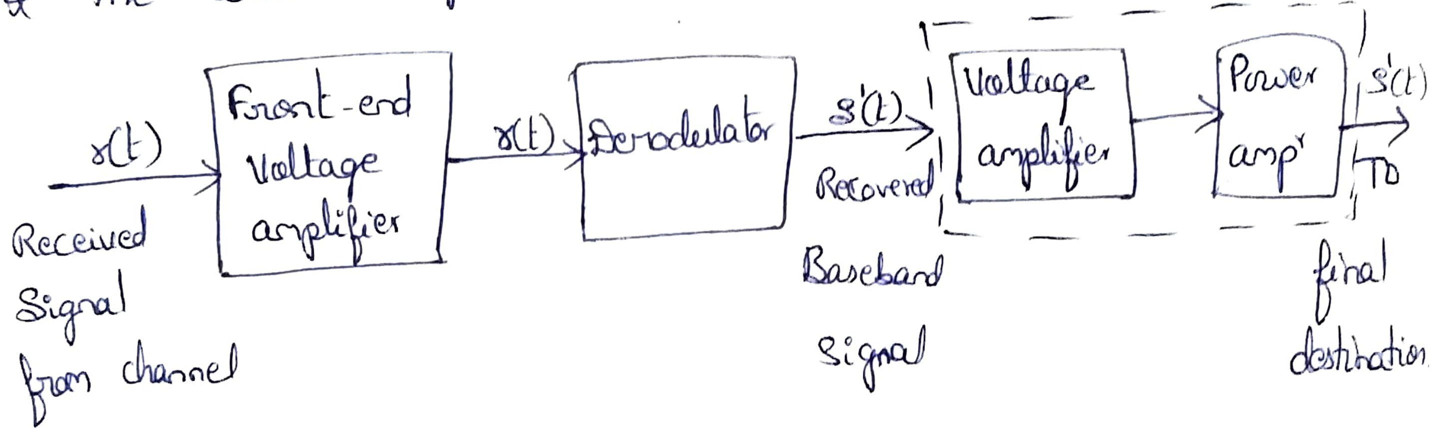
→ First, dc power supply made zero, which makes diode current as zero & which Switch off the diode. At that, time measure the o/p Power.

→ Now, apply dc power supply & the diode turns ON, which results in diode current & adjust the i/p to any value, measure the o/p power.

$$\boxed{F = 19.3\, R_i I_S}$$ //.

RECEIVER :- (RXR)

* After transmitting through the channel, the Signal is reaching to RXR.
* RXR performs reverse operation of the TXR, to get original baseband Signal.
* The block diagram of the receiver is shown below.

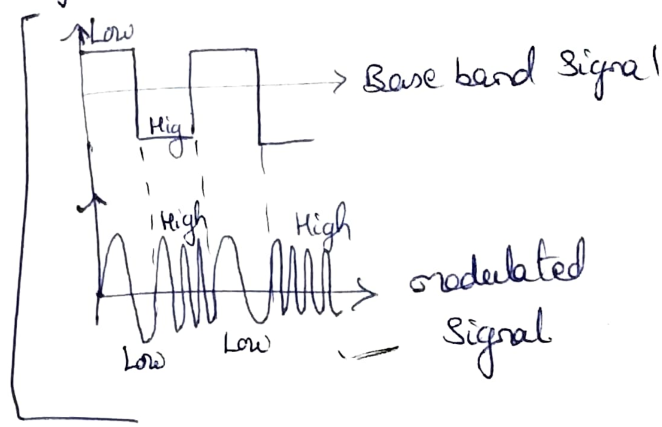


→ The operation of the RXR is to remove the noise, which added during the transmission.

→ Since, RXR receives weak Signal as the transmitted Signal loose its strength during transmitting in the channel, hence RXR first amplifies the received Signal by using voltage amplifier (front end) Then recovering of the original baseband signal can be done.

→ The pre-amplified received signal is demodulated to convert high-freq spectrum signal to original low-freq spectrum signal. This happens in carrier communication s/m.

→ But, in Baseband communication s/m, as TXR replaces Digital baseband signal into a series of two difnt freq sine waves,



the RXR receives this signal & it recovers the original baseband pulse by replacing two sine waves with the corresponding original signal.

∝ Now, recovered Baseband signal is handed over to the O/P transducer, which converts electrical energy to its actual form of energy, then applied to final destination.

* The final destination may be a user (or) a machine. For Ex: in Display machines like TV. The TV RXR receives the signal & converts those into actual Videos & display those onto the monitor.

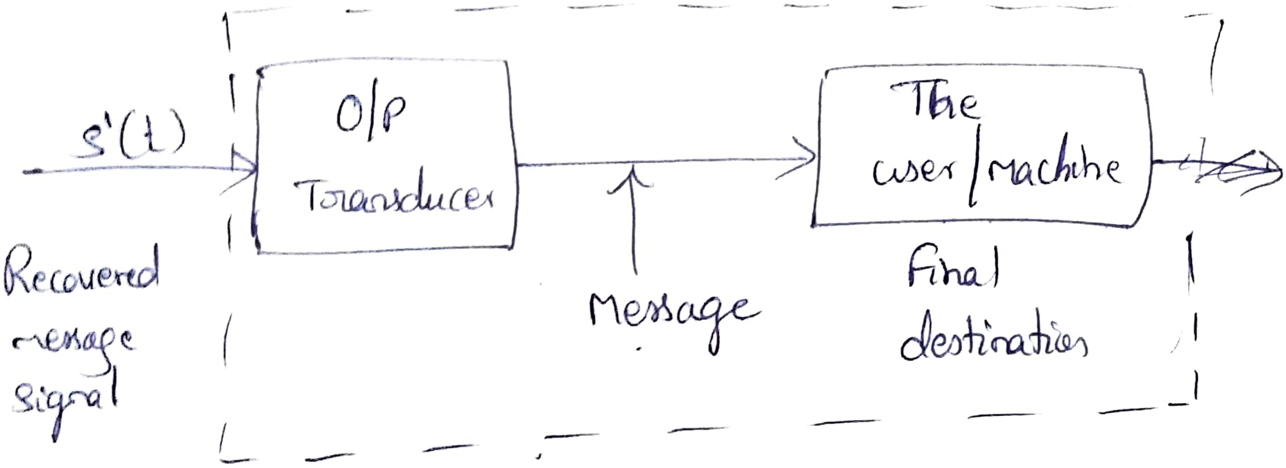


figure: Block diagram of the final destination.

Indetailed Block Diagram of Basic Communication System: —

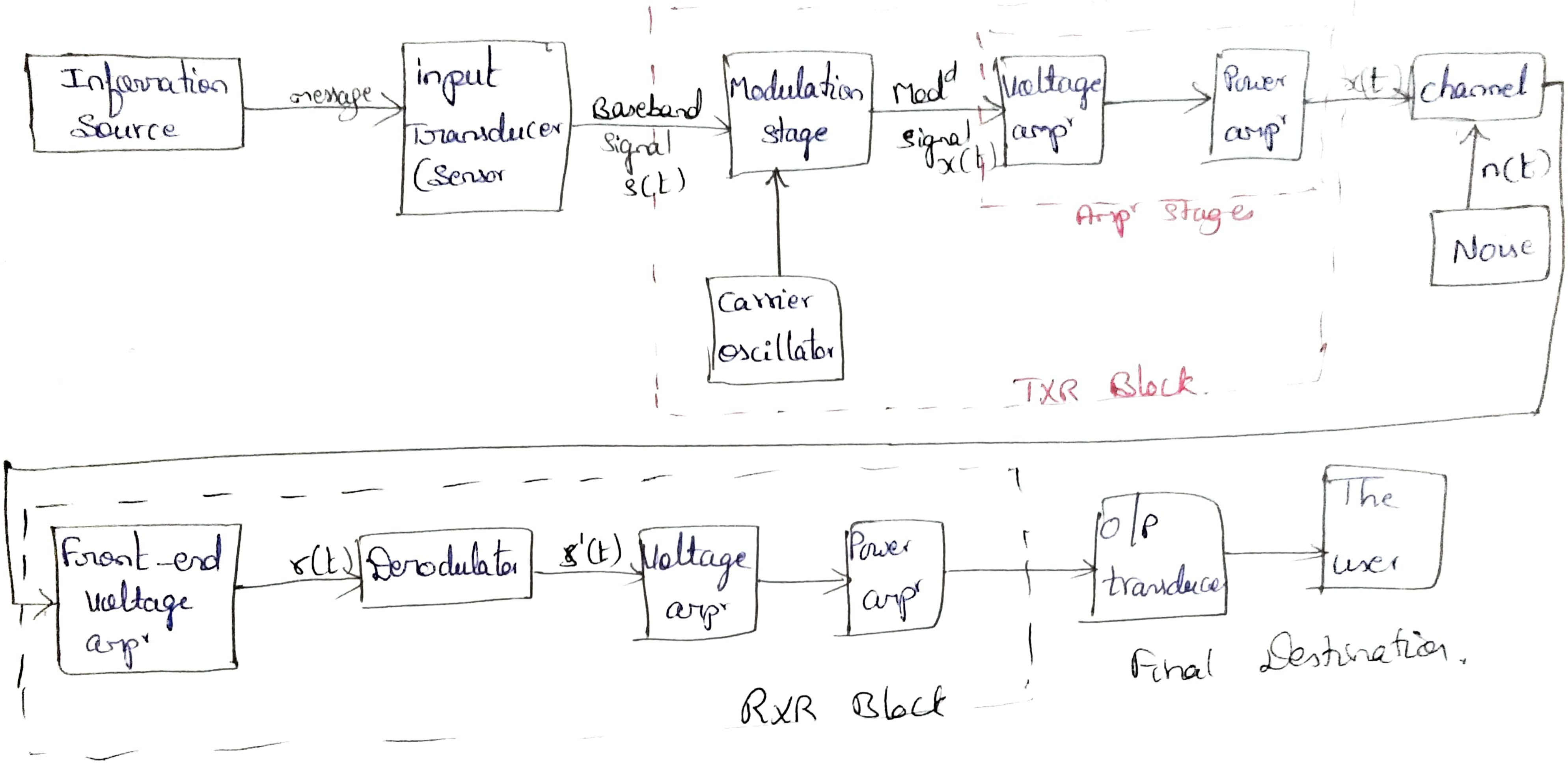


figure: Detailed Block Diagram of Communication System.

MULTIPLEXING:—

* Multiplexer is a logic circuit, which selects one of the input as its o/p from multiple number of i/p lines.
* In the communication s/m, it plays an important role.
* If 10 persons starts to speaks at the same time in a meeting (or) anywhere, then no ones voice we can recognize properly. It becomes a noise, for our ear & our brain is not having such intelligence.

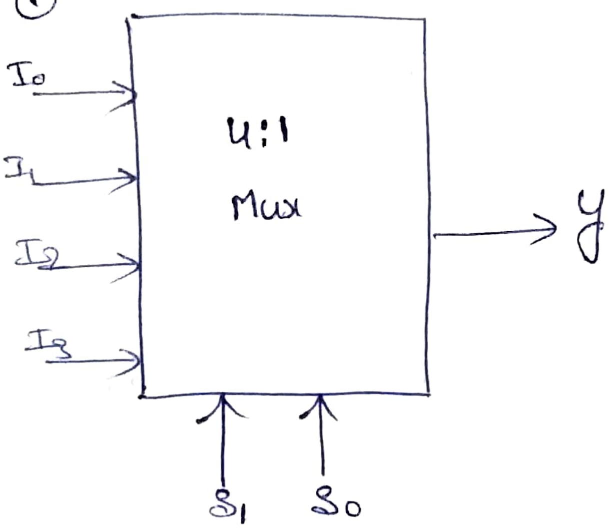
→ In the similar manner, in a communication s/m if more than one signal is transmitting at the same time, then the RXR becomes very difficult to identify the signals individually.

→ Hence, Multiplexing process makes it easier to identify & receive the signals individually.

→ Multiplexing selects only one i/p signal as its o/p based on the requirement mentioned in the select lines.

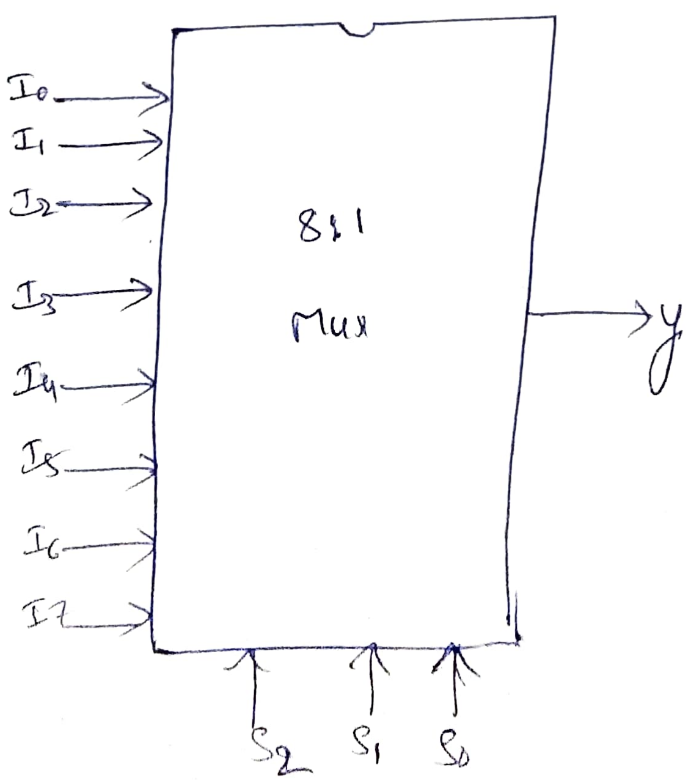
→ In Multiplexing process, each of the signal is transmitted with a dif^nt frequency spectrum (by modulation) so that each signal is easily distinguishable from the other (even they are within the same audio band).

Ex: ①



inputs		o/p
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

②



inputs			o/p
S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

* In multiplexer,

For | 'n' no^r of Select lines
$2^n \rightarrow$ ______ input lines
$1 \rightarrow$ o/p lines
//

TYPES OF COMMUNICATION S/M

* Communication Systems are of dif^nt types based on:
→ Physical infrastructure,
→ Specifications of the signals they transmit.

① **Based on the physical infrastructure:**

* Based on the physical link between TXR & RXR, the communication Systems are of two types. They are:

→ Line or wired or H/w communication.

→ Radio or wireless or S/w ______

* In Line / wired / H/w communication, there is a physical link exist in b/w TXR & RXR.

  Ex: Land line telephony

* In wireless / Radio / s/w communication, there is no such physical link exists in b/w TXR & RXR.

  Ex: Satellite communication.

* Depends on the Txn of the signal, the communication is divided as 3-types. They are:
  - Simplex,
  - Duplex,
  - Half duplex.

⇒ Simplex ⇒ "One way Transmission".

  Ex: A TV RXR can receive the signals, but can't transmits.

⇒ Duplex ⇒ "Two way Txn" (One can simultaneously send & receive the signals).

  Ex: communication through mobile.

⇒ Half Duplex ⇒ "Two way Txn" (Can be txd or Rxd at a time)

  [But can't received & Txt at a time]

② Based on Specifications of the Signal they transmit :-

* Types of communication include :
→ Nature of the Baseband Signal,
→ ——〃—— Transmitted Signal.

* As the Baseband Signal may be analog (or) digital, based on that there are two types of communication s/ms.

They are ;
→ Analog communication S/m, Ex: TV
→ Digital ——〃—— . Ex: HDTV.

* As the transmitted signal modulated during the transmission in the TXR block, based on that there are two types of communication s/ms again. They are :
→ Baseband Communication S/m, Ex: Land line Telephony. Fax
→ Carrier ——〃—— . Ex: Radio broadcast, Cable TV,

* Finally, a communication s/m may be any one of the 8-types, which makes as 4-groups.

They are:

→ Group I : Line / Radio communication s/m,
→ Group II : Simplex / Duplex ——"——,
→ Group III : Analog / Digital ——"——,
→ Group IV : Baseband / carrier ——"——.

Ex: ① TV Communication s/m is referred as:

$\boxed{\text{Radio – Simplex – Analog – Carrier communication s/m}}$

② Land line Telephony is:

$\boxed{\text{Line – Duplex – Analog – Baseband comm}^{n}\text{ s/m}}$

③ Mobile comm^{n} is

$\boxed{\text{Radio – Duplex – Digital – Carrier comm}^{n}\text{ s/m}}$

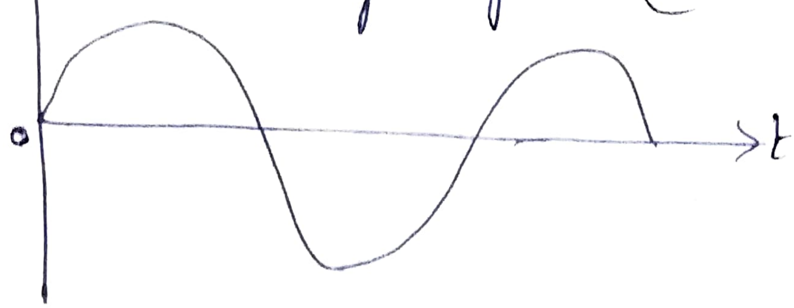
Modulation:

* It is a process of converting low frequency spectrum signal into High frequency spectrum.
* There are dif^nt types of modulations. They are:
  - Amplitude modulation [AM],
  - Frequency modulation [FM],
  - Phase —— " —— [PM],
    - Pulse width modulation (PWM),
    - Pulse Position —— " —— (PPM),
    - " Code —— " —— (PCM).

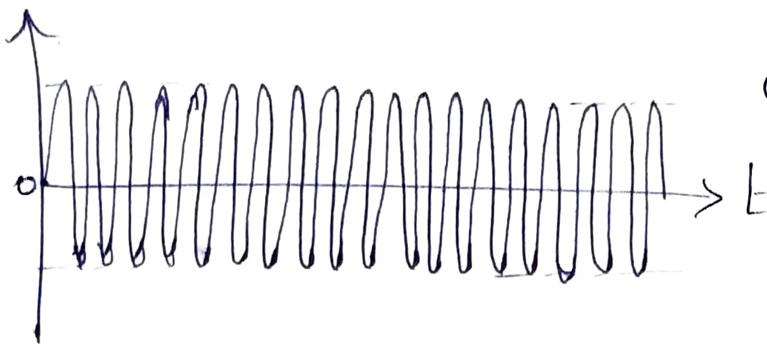
Amplitude Modulation: — [AM]

* AM is a modulation technique, in which instantaneous amplitude of a Carrier Signal is varied in accordance with the instantaneous amplitude of the analog modulating Signal to be transmitted.
* The modulating Signal is an analog based Signal with low frequency (as shown below).

Modulating Signal (Has low freq)

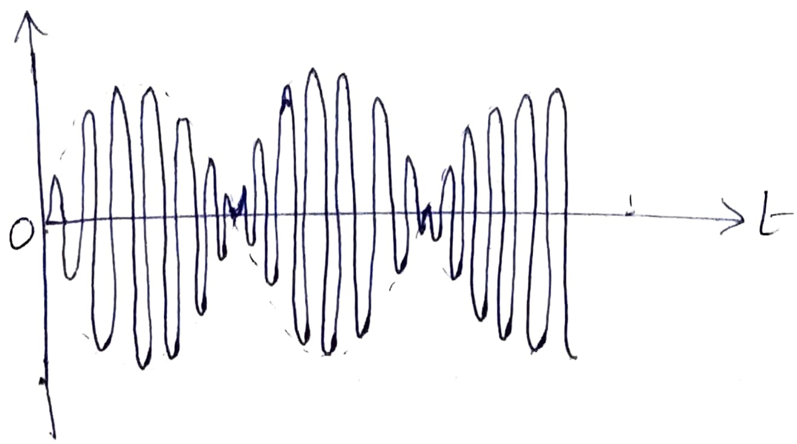


- Carrier Signal is always a <u>Sine wave</u> with High frequency as shown below.



Carrier Signal.

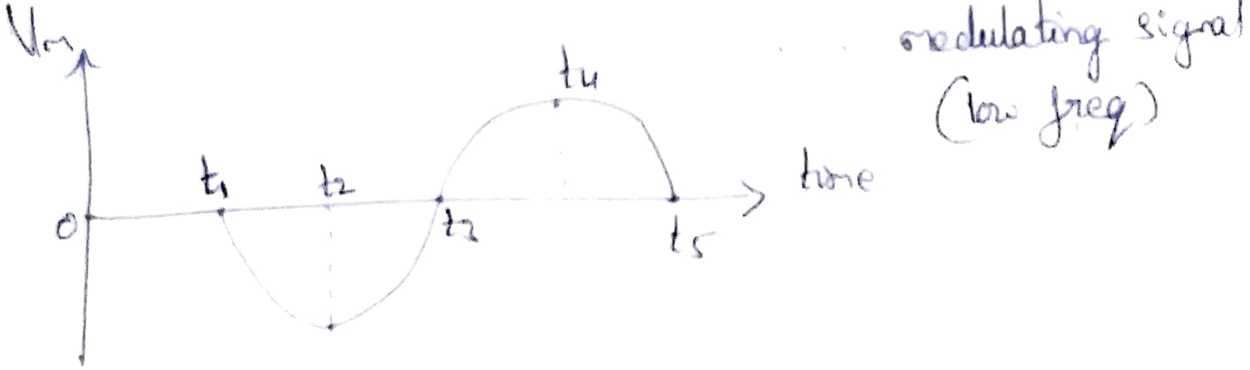
- Modulated Signal can be as follows.



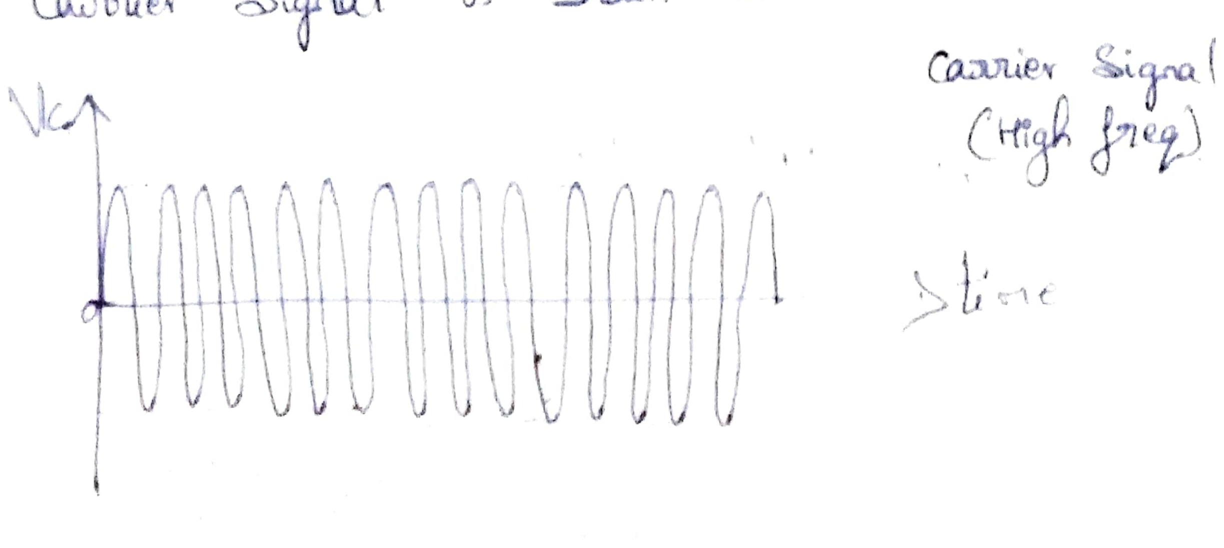
FREQUENCY MODULATION: — [FM]

* FM is a type of modulation technique, in which instantaneous frequency of the carrier signal is changed according to the instantaneous amplitude of the modulating (or) message signal.

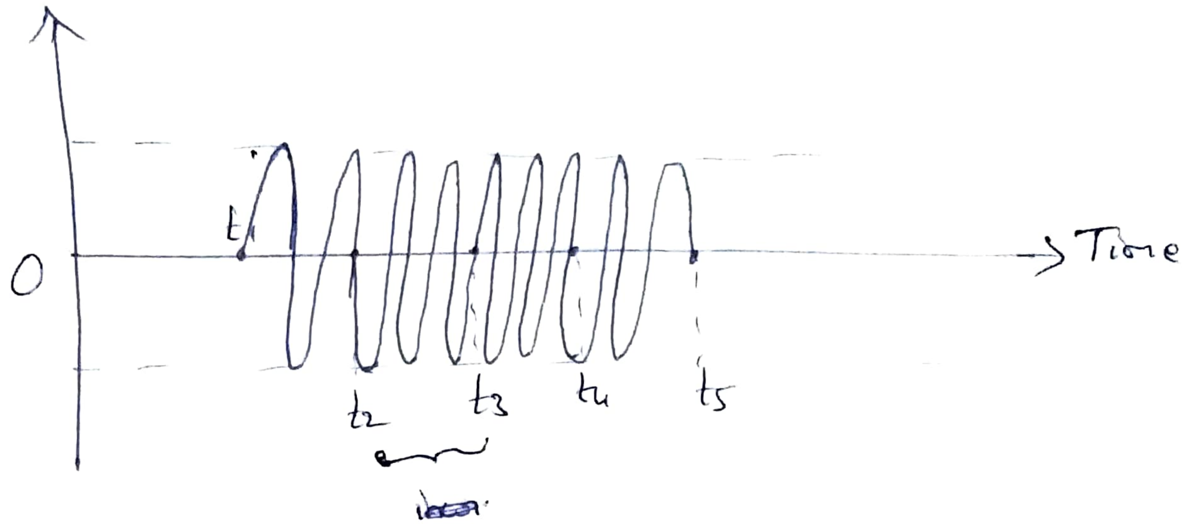
* consider a modulating signal (or signal to be transmitted for modulation) as shown below.



* Carrier Signal is shown below.



* Freq modulated signal is shown below.



From t_1 to t_2 interval $\Rightarrow$ freq decreases,

From t_2 to t_3 interval $\Rightarrow$ freq slightly increases,

From t_3 to t_4 — $\Rightarrow$ — rappidly —,

From t_4 to t_5 — $\Rightarrow$ — slightly decreases.

RADIO WAVES :—

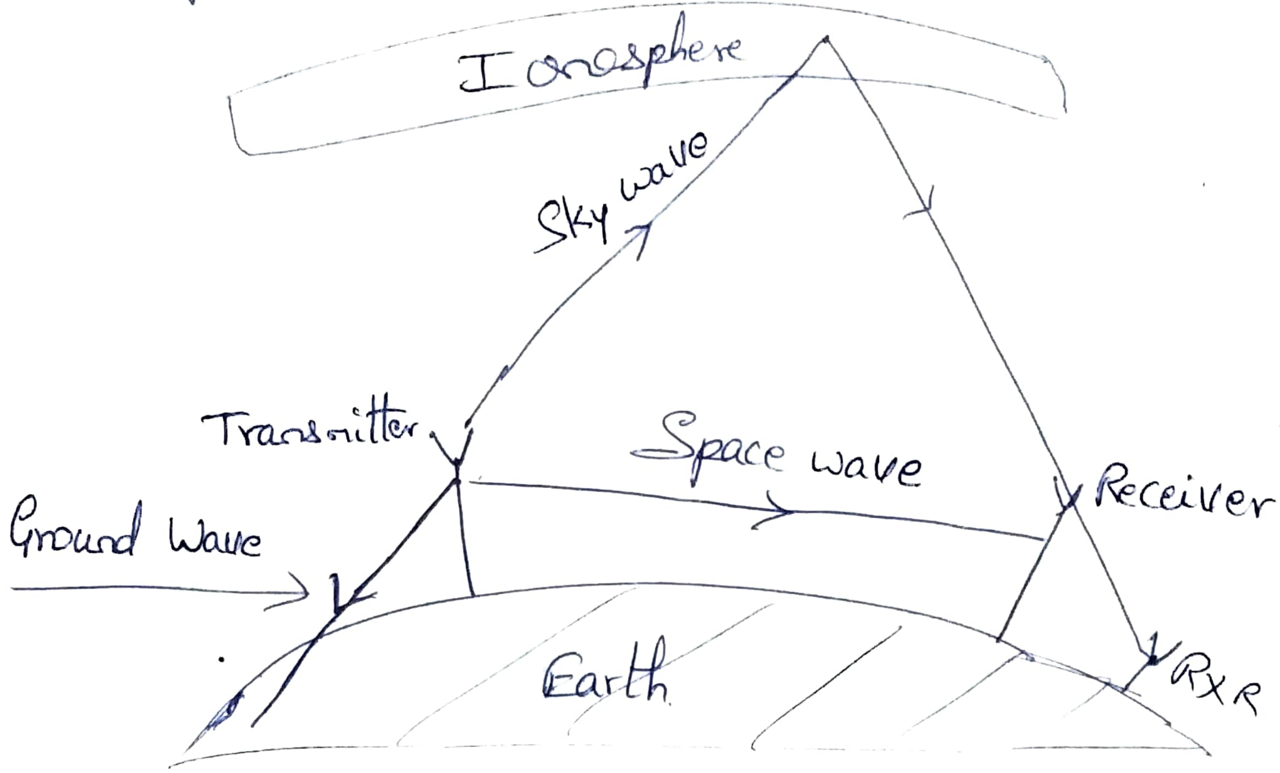
In Space Communication, the Electromagnetic waves (EM Waves) of dif^nt^ frequencies (10 kHz to 300 GHz) are referred as "Radio Waves".

- These radio waves travels from transmitting antenna to receiving antenna in several ways. On the basis of mode of propagation, the radio waves are broadly classified as :

  1. Ground or Surface wave,
  2. Space or Tropospheric wave,
  3. Sky wave.

- According to dif^nt^ types of waves, there are 3-types of propagations. They are,
  - → Ground Wave propagation,
  - → Space or Tropospheric wave propagation,
  - → Sky Wave propagation.

* Below figure shows, all the 3-modes of radio wave propagation.



① Ground (or) Surface Wave Propagation :-

* In Ground wave propagation, the radio waves are travelling along the surface of the earth.

* The high frequency waves are absorbed by the earth, hence low frequency waves are guided by the earth & hence ~~low~~ Ground wave propagation is useful at

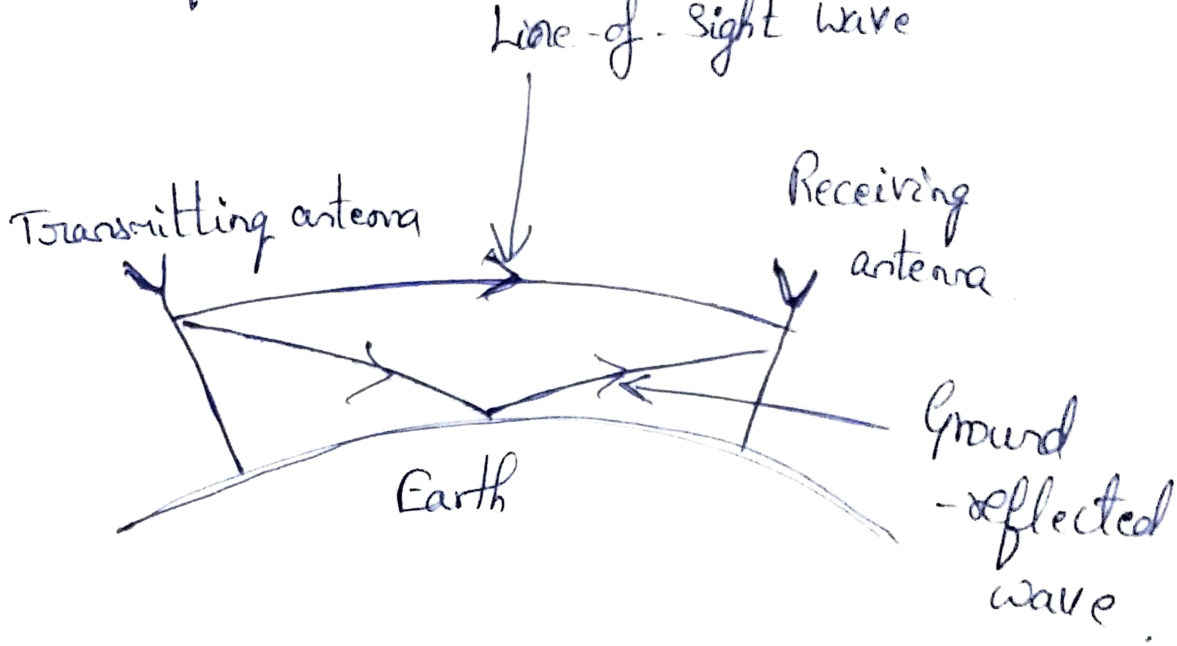
low frequencies.

* Ground waves are used for communication at frequency below 500 kHz within a distance of 1500 km from the transmitter.

(8) Space (or) Tropospheric Wave propagation :-

* The radio waves which travel at a distance of 15 km from the surface of the earth, are referred as Space Waves.
* These Space waves always b/w transmitting antenna & Receiving antenna.
* They can travel in two ways,
  → Directly from Transmitting antenna to receiving antenna, which is referred as Line-of-Sight propagation.

* They may reflect from the ground & then reaches to receiving antenna. Such radio waves are referred as Ground-Reflected Waves.



(3) Sky Wave Propagation :—

* In this mode of propagation, the radio waves reflects from the ionosphere in the sky & reflected back to the earth at the receiving antenna.